



第十三届物流与海事系统国际会议

13th International Conference on Logistics and Maritime Systems

Program Book

Shanghai, China | 2025年8月



上海海事大学
SHANGHAI MARITIME UNIVERSITY



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LOGMS 2025

第十三届物流与海事系统国际会议

The 13th International Conference on
Logistics & Maritime Systems

***Systematic and Sustainable Evolution of Maritime Logistics:
Green, Intelligent, Multimodal, and Port-City Synergy***

August 26-29, 2025

Shanghai, China

<https://logms2025.shmtu.edu.cn>

Welcome address by SMU

Distinguished guests, ladies and gentlemen,

Good morning!

On behalf of the host institution, Shanghai Maritime University, I would like to extend my warmest congratulations on the successful opening of the 13th International Conference on Logistics and Maritime Systems (LOGMS 2025), and to offer my most sincere welcome to all experts, scholars, and industry colleagues who have traveled from near and far to join us today.

With the theme “*Systematic and Sustainable Evolution of Maritime Logistics: Green, Intelligent, Multimodal, and Port-City Synergy*”, this conference seeks to gather wisdom from around the world to exchange frontier ideas, showcase innovative technologies, and explore future pathways.

Since its inception in 2005, the LOGMS conference series has grown into an influential academic forum in the fields of logistics and maritime systems. In 2008, Shanghai Maritime University (SMU) hosted the 4th LOGMS. Seventeen years later, it is our great honor to once again welcome LOGMS back to our campus. As a multidisciplinary research-oriented university with distinctive strengths in shipping, logistics, and marine studies, SMU has established a comprehensive academic structure. The University has long been committed to advancing scientific innovation and talent cultivation in shipping and logistics. Building on our traditional strengths in port management and international transport, we continue to expand research in areas such as smart ports, green shipping, supply chain collaboration, multimodal transport optimization, and port-city integration.

We hope that through this high-level international platform, all participants will engage in meaningful exchanges, inspire new ideas, and foster closer cooperation, contributing together to a smarter, greener, and more integrated future for maritime logistics.

Finally, I wish LOGMS 2025 great success and sincerely hope that all participants enjoy fruitful exchanges and a rewarding experience during your stay in Shanghai.



YAN Wei

Vice President, Shanghai Maritime University
Shanghai, China



Welcome Address by LOGMS2025

Dear Distinguished Guests, Speakers, and Participants,

We are delighted to welcome you to the 13th International Conference on Logistics and Maritime Systems (LOGMS 2025), which will be held from August 26 to 29, 2025, in Shanghai, China.

This conference offers a dynamic platform for professionals, researchers, and scholars to engage in in-depth discussions on key topics such as management science and engineering, green logistics, energy efficiency, optimization of collection and distribution systems, and port-city coordination within the broader context of intelligent and sustainable development.

Since its inception in 2005, the LOGMS conference series has served as a significant forum for advancing knowledge and innovation in the field of logistics and maritime systems. In 2008, Shanghai Maritime University (SMU) proudly hosted the conference, and it is our honor to once again serve as the host institution for LOGMS 2025, continuing our commitment to supporting excellence in maritime research and collaboration.

We believe that your insightful presentations, thoughtful questions, and engaging discussions will contribute meaningfully to the advancement of our field. We warmly welcome you to LOGMS 2025 and sincerely hope that your participation will be both intellectually rewarding and personally memorable, fostering new connections and lasting partnerships.

Below are the conference details.

The theme of the Conference

*Systematic and Sustainable Evolution of Maritime Logistics:
Green, Intelligent, Multimodal, and Port-City Synergy.*

Conference Date & Venue

Date: August 26-29, 2025
Venue: Shipping Science Building, Shanghai Maritime University, Shanghai, China.
Address: No. 1550 Haigang Avenue, Pudong New Area, Shanghai. (Near Gate 5)
上海海事大学 航运科技创新大楼（靠近 5 号门）

Conference Organization

Hosted by:

Shanghai Maritime University

Organized by:

Institute of Logistics Science and Engineering, Shanghai Maritime University

Co-organizers:

Container Supply Chain Engineering Research Center, Ministry of Education
Key Laboratory of Transport Industry of Marine Technology and Control Engineering
Research Sub-center of Multimodal Transport Center for Yangtze River Economic Belt
Shanghai Transportation Development Research Center
Shanghai Miluo Information Technology Co., Ltd.



Conference Committee

Chair



YAN Wei

Vice President

Shanghai Maritime University

Vice Chair



SUN Shibin

Dean of Institute of Logistics Science and Engineering

Shanghai Maritime University

Organization Chair



LIANG Chengji

Professor

Shanghai Maritime University



DING Yi

Professor

Shanghai Maritime University

Organization Committee



WANG Yu

Shanghai Maritime University



ZHOU Yanjie

Zhengzhou University



Conference Secretary

ZHAO Rui, Shanghai Maritime University

ZHANG Zicheng, Shanghai Maritime University

CHEN Shiqi, Shanghai Maritime University

XU Xiaojie, Shanghai Maritime University

Steering Committee

Rommert DEKKER	Erasmus University Rotterdam, Netherlands
Kaphwan KIM	Zhejiang University, China
Hyerim BAE	Pusan National University, Korea
Paul Tae-Woo LEE	Zhejiang University, China
Chun-Hsien CHEN	Nanyang Technological University, Singapore
GE Ying-en	Chang'an University, China
Haobin LI	National University of Singapore, Singapore
Stefan VOSS	University of Hamburg, Germany
LIANG Chengji	Shanghai Maritime University, China
JIN Jiangang	Shanghai Jiao Tong University, China
Mitsuo GEN	Tokyo University of Science, Japan
YANG Bin	Shanghai Maritime University, China
Il-Kyeong MOOM	Seoul National University, Korea
Xiuju FU	Institute of High Performance Computing, Agency for Science, Technology and Research (A*STAR), Singapore
TANG Xuhong	Shanghai Maritime University, China
ZHOU Yanjie	Zhengzhou University, China
HE Junliang	Shanghai Maritime University, China
HAN Hao	Shanghai Maritime University, China
FENG Xuehao	Zhejiang University, China
DING Yi	Shanghai Maritime University, China
LIN Lin	Dalian University of Technology, China
JIA Shuai	Hong Kong University of Science and Technology, China

Keynote & Invited Talks

Prof. Rommert DEKKER, Erasmus University Rotterdam, Netherlands:

“Ports on their way to the future: examples from Rotterdam”

Prof. GE Ying-en, Chang'an University, China:

“Optimizing liner shipping routes and speed profiles considering sulfur emission control zones and carbon taxes”

Prof. Stefan VOSS, University of Hamburg, Germany:

“Digital twins in seaports and container terminals”

Dr. Haobin LI, National University of Singapore, Singapore:

“Beyond the data: rethinking AI applications for logistics and maritime systems”

Dr. Xiuju FU, Institute of High Performance Computing, Agency for Science, Technology and Research (A*STAR), Singapore:

“Navigating the future of shipping: AI innovations and their impact”

Prof. Hyerim BAE, Pusan National University, Korea:

“AI-powered smart port logistics: lessons from South Korea’s national projects”

Prof. Mitsuo GEN, Tokyo Univ. of Science, Japan:

“Recent advances in multiobjective sustainable closed-loop SCM network models”

FANG Huaijin, Shanghai International Port (Group) Co., Ltd. (SIPG), China:

“Green port construction and prospects”

WANG Beili, Shanghai E&P International, Inc., China:

“From Practice to Foresight: Smart Innovation in Intelligent Logistics and the Digital Development Path of Shipping & Trade”

Prof. YANG Bin, Shanghai Maritime University, China:

“Practical challenges and reflections on multimodal transport development in the Yangtze River Economic Belt”

Prof. Paul Tae-Woo LEE (Korea), Zhejiang University, China:

“Navigating the high seas: reshaping global trade and supply chains in a US tariff-driven world”

Prof. Chun-Hsien CHEN, Nanyang Technological University, Singapore:

“AI-supported human factors in ATM and VTM research”

Prof. FENG Xuehao, Zhejiang University, China:

“Development of green maritime logistics industry: Coordinated policy making and interdisciplinary studies”

Prof. JIA Shuai, The Hong Kong University of Science and Technology, China:

“An integrated framework for truck arrival management and yard capacity utilization in a container terminal”

Prof. DING Yi, Shanghai Maritime University, China:

“Lecture on artificial intelligence and digital port logistics”

Prof. LIANG Chengji, Shanghai Maritime University, China:

“Innovative models in port collection-distribution systems — underground logistics applications”



Prof. JIN Jiangang, Shanghai Jiao Tong University, China:

“Strategic planning for containership green transition under net-zero emission target: a predict-then-optimize framework”

Prof. WANG Tingsong, Shanghai University, China:

“Column generation approach for the problem of integrated operations in a container terminal under carbon emission tax”

Prof. XIAO Guangnian, Shanghai Maritime University, China:

“The new pattern of global shipping carbon emissions under the influence of the IMO's new decarbonization regulations and the EU emissions trading system”

Prof. HE Junliang, Shanghai Maritime University, China:

“AI-driven carbon peaking optimization for container intermodal transport networks”

Program & Schedule

Conference Program:

		Shanghai Maritime University, Shipping Science Building 上海海事大学 航运科技创新大楼 No.1550 Haigang Avenue,Pudong New Area, Shanghai.(Near Gate 5)					
		B301	B303	B304	B305	Lobby	
Aug.26 (Tues.)	13:00-20:00	Registration				Registration	
Aug. 27 (Wed.)	09:00-09:30	Opening Greetings					
	09:30-10:30	Keynote Talk I					
	10:30-11:00	Group Photo & Coffee Break (The Third Floor Lobby)					
	11:00-12:00	Keynote Talk II					
	12:00-14:00	Lunch (The Third Floor Lobby)					
	14:00-15:30	Keynote Talk III					
	15:30-15:50	Coffee Break (The Third Floor Lobby)					
	15:50-17:40	Invited Talk I	Session 1 Port Safety and Detection	Session 2 Vessel Traffic and Scheduling	Session 3 Intelligent Optimization of Yard & Transportation		
	18:30-20:30	Dinner Buffet (The Third Floor Lobby)					
Aug. 28 (Thu.)	09:00-10:30	Invited Talk II	Session 4 Integrated Optimization of Container Terminal	Session 5 Automated Terminal Smart Operations	10:00-12:00 Invited Session Multi-modal Terminals and Transportation:Data driven approach		
	10:30-10:50	Coffee Break (The Third Floor Lobby)					
	10:50-12:20	Invited Talk IV	Session 6 New Perspectives on Shipping Policy	Session 7 Machine Learning Applications			
	12:20-14:00	Lunch (The Third Floor Lobby)					
	14:00-15:30	Best Paper Competition	Session 8 Maritime Supply Chain	Session 9 Port Energy Supply & Strategies	Multi-modal Terminals and Transportation(I)		
	15:30-15:50	Coffee Break (The Third Floor Lobby)					
	15:50-17:40	LOGMS Committee Meeting	Session 10 Shipping Network & Digitalization	Session 11 Logistics VRP Methods	Multi-modal Terminals and Transportation(II)		
		18:30-20:30	Closing & Awarding (Radisson Collection Lingang Shanghai)				
Aug. 29 (Fri.)	9:00-12:00	Visit : Yangshan Automated Terminal					

No.100 Dingke Road,Pudong New Area
上海临港国际酒店

Radisson Collection Lingang Shanghai
上海临港锦江国际酒店
No.100 Dingke Road, Pudong New Area



LOGMS 2025 SCHEDULE

Shanghai Maritime University, Shipping Science Building
上海海事大学 航运科技创新大楼（近 5 号门）
No.1550 Haigang Avenue,Pudong New Area, Shanghai. (Near Gate 5)

August 27 (Wednesday): Shipping Science Building B301

08:30-09:00

Registration

Opening Greetings

Host: Prof. SUN Shibin

Opening Remarks: Prof. SUN Shibin, Dean of the Institute of Logistics Science and Engineering

09:00-09:30

Welcome Address: Prof. YAN Wei, Vice President of Shanghai Maritime University

Guest Remarks: Prof. Rommert DEKKER, Steering Committee Chair of LOGMS

Conference Overview

Keynote Talk Part I

Chair: Prof. Paul Tae-Woo LEE

09:30-10:00

Keynote Presentation 1: *Ports on their way to the future: examples from Rotterdam*

Prof. Rommert DEKKER, Erasmus University Erasmus, Netherlands

10:00-10:30

Keynote Presentation 2: *Optimizing liner shipping routes and speed profiles considering sulfur emission control zones and carbon taxes*

Prof. GE Ying-en, Chang'an University, China

10:30-11:00

Group Photo & Coffee Break (3rd Floor Lobby)

Keynote Talk Part II

Chair: Prof. Rommert DEKKER

11:00-11:30

Keynote Presentation 3: *Digital Twins in Seaports and Container Terminals*

Prof. Stefan VOSS, University of Hamburg, Germany

11:30-12:00

Keynote Presentation 4: *Beyond the Data: Rethinking AI Applications for Logistics and Maritime Systems*

Prof. Haobin LI, National University of Singapore, Singapore

12:00-14:00

Lunch (3rd Floor Lobby)

Keynote Talk Part III

Chair: Prof. DING Yi

14:00-14:30

Keynote Presentation 5: *Navigating the Future of Shipping: AI Innovations and Their Impact*

Prof. Xiuju FU, Institute of High Performance Computing, Agency for Science, Technology and Research (A*STAR), Singapore.

14:30-15:00

Keynote Presentation 6: *AI-Powered Smart Port Logistics: Lessons from South Korea's National Projects*

Prof. Hyerim BAE, Pusan National University, Korea.



15:00-15:30	Keynote Presentation 7: <i>Recent Advances in Multiobjective Sustainable Closed-loop SCM Network Models</i> Prof. Mitsuo GEN, Tokyo Univ. of Science, Japan.
15:30-15:50	Coffee Break (3rd Floor Lobby)
Invited Talk Part I Chair: Prof. LIANG Chengji	
15:50-16:20	Invited Talk 1: <i>Green Port Construction and Prospects</i> FANG Huaijin, Shanghai International Port (Group) Co., Ltd. (SIPG), China.
16:20-16:50	Invited Talk 2: <i>From Practice to Foresight: Smart Innovation in Intelligent Logistics and the Digital Development Path of Shipping & Trade</i> WANG Beili, Shanghai E&P International, Inc., China
16:50-17:20	Invited Talk 3: <i>Practical Challenges and Reflections on Multimodal Transport Development in the Yangtze River Economic Belt</i> Prof. YANG Bin, Shanghai Maritime University, China.
18:30-20:30	Dinner Buffet (3rd Floor Lobby)

August 27 (Wednesday): Shipping Science Building B303

Parallel Session 1: Port Safety and Detection Chair: Dr. FU Guanghua	
	1. <i>Learning-based Assessment for Port Occupational Safety</i> Sang Duck LEE (Korea Railroad Research Institute)
	2. <i>Multimodal Sensor Projection and Feature-Based Clustering for Abnormal Vessel Detection</i> Kyunghun LEE (Pusan National University)
	3. <i>A Dynamic Stochastic Optimization Framework for De-risking Green Port Investments: The Case of Shore Power in Guangzhou</i> GUO Yanru (Guangzhou Maritime University)
15:50-17:50	4. <i>Drone Scheduling Optimization for Continuous Sea Area Emission Monitoring</i> LIU Yun (Shanghai Jiao Tong University)
	5. <i>An Analogical Corpus-Based Study of Maritime Proposal Convergence Trends for Chinese and English Native Speakers</i> DU Juan (Shanghai Maritime University)
	6. <i>Carbon Emission Reduction Potential of Express Delivery Transportation under Road-to-Rail Policy</i> CHEN Shiqi (Shanghai Maritime University)
18:30-20:30	Dinner Buffet (3rd Floor Lobby)



August 27 (Wednesday): Shipping Science Building B304

Parallel Session 2: Vessel Traffic and Scheduling

Chair: Prof. CHEN Xinqiang

- 15:50-17:50
1. *What Weather is Good Weather under Performance Clauses in Charterparties?*
Haiying JIA (Norwegian School of Economics)
 2. *An Investigation into the Optimization of Vessel Unloading Scheduling for Triple Hoist Quay Cranes in Two-Story Container Terminals*
HU Hongtao (Shanghai Maritime University)
 3. *A Rolling-Horizon Model for Fair and Efficient Just-In-Time Arrival of Ships*
Reinier VAN UDEN (Erasmus University Rotterdam)
 4. *Joint vessel deployment strategy considering individual operational preferences*
WANG Xudong (Shanghai Maritime University)
 5. *How Reservation and Customer Ranking Mechanism Affect Efficiency, Revenue and Fairness in the Panama Canal's Transits?*
MA Zhongjun (Tsinghua University)
 6. *Reinforcement Learning-Based Dynamic Slot Allocation In Container Yards Considering Container Cluster Strategy*
TANG Dong (Shanghai Maritime University)

18:30-20:30

Dinner Buffet (3rd Floor Lobby)

August 27 (Wednesday): Shipping Science Building B305

Parallel Session 3: Intelligent Optimization of Yard & Transportation

Chair: Dr. LI Yi

- 15:50-17:50
1. *Mixed Car-Carrier Fleets Capacity Design for Finished-Vehicle Logistics: A Column Generation Method*
QU Weijia (Shanghai Maritime University)
 2. *Ocean Container Shipping Pricing Considering Empty Container Repositioning and Containerized Bulk Cargo Transportation*
YU Mingzhu (Shenzhen University)
 3. *Study on coordination mechanism and pricing of lithium carbonate option contract in closed-loop power battery supply chain under supply constraints in recycling market*
WEI Hairui (University of Shanghai for Science & Technology)
 4. *A data-driven rolling-forecast-horizon LSTM-based Wagner-Whitin algorithm for establishing a smart coal port inventory system under uncertainties*
DING Peipei (Fuzhou University)



5. *Optimization of Multi-Energy Rubber-Tired Crane Scheduling under Uncertainty*

GAO Yinping (Shanghai Maritime University)

6. *Intelligent Positioning Method for Ship Block Yard Based on Spatial-Temporal Bi-Objective Constraints*

DONG Xinxin (Shanghai Maritime University)

18:30-20:30

Dinner Buffet (3rd Floor Lobby)

August 28 (Thursday) : Shipping Science Building B301

Invited Talk Part II

Chair: Prof. JIA Shuai

09:00-09:30 **Invited Talk 4:** *Navigating the High Seas: Reshaping Global Trade and Supply Chains in a US Tariff-Driven World*

Prof. Paul Tae-Woo LEE (Korea), Zhejiang University, China

09:30-10:00 **Invited Talk 5:** *AI-supported Human Factors in ATM and VTM Research*
Prof. CHEN Chun-Hsien, Nanyang Technological University, Singapore.

10:00-10:30 **Invited Talk 6:** *Development of green maritime logistics industry: Coordinated policy making and interdisciplinary studies*
Prof. FENG Xuehao, Zhejiang University, China

10:30-10:50

Coffee Break (3rd Floor Lobby)

Invited Talk Part IV

Chair: Prof. HU Hongtao

10:50-11:20 **Invited Talk 7:** *An integrated framework for truck arrival management and yard capacity utilization in a container terminal*

Prof. JIA Shuai, The Hong Kong University of Science and Technology

11:20-11:50 **Invited Talk 8:** *Lecture on artificial intelligence and digital port logistics*
Prof. DING Yi, Shanghai Maritime University, China.

11:50-12:20 **Invited Talk 9:** *Innovative Models in Port Collection-Distribution Systems—Underground Logistics Applications*
Prof. LIANG Chengji, Shanghai Maritime University, China.

12:20-14:00

Lunch (3rd Floor Lobby)

Best Paper Competition

Chair: Prof. Rommert DEKKER, Prof. Tae-Woo LEE, Stefan VOSS; Assistant: Prof. ZHOU Yanjie

14:00-15:30

Best Paper Competition

15:30-15:50

Coffee Break (3rd Floor Lobby)

LOGMS Committee Meeting

Assistant: Prof. ZHOU Yanjie

15:50-17:20

LOGMS Committee Meeting

18:30-20:30

Dinner & Banquet (Radisson Collection Lingang Shanghai)



August 28 (Thursday) : Shipping Science Building B303

Parallel Session 4: Integrated Optimization of Container Terminal

Chair: Prof. CHEN

Xinqiang

- 09:00-10:30
1. *Robust vessel traffic scheduling with uncertain berth service times in a seaport*
ZHAO Runqing (The Hong Kong University of Science and Technology (Guangzhou))
 2. *Berth and quay cranes allocation problem with on-shore power supply assignment in container terminals*
WANG Zhu (Shanghai Maritime University)
 3. *Cost-Efficient Quay Crane Assignment and Scheduling Under Time-Variant Surcharge Policies in Container Terminals*
Minsu WOO (Dankook University)
 4. *Stack-Based Yard Allocation Considering Inter-Terminal Transportation in a Transshipment Hub*
Seongbae JO (Seoul National University)
 5. *Activity-based or Behaviour-based? A Comparative Evaluation of Hybrid Modeling Formalism for Container Terminal Operations*
SHAO Changying (National University of Singapore)

10:30-10:50

Coffee Break (3rd Floor Lobby)

Parallel Session 6: New Perspectives on Shipping Policy

Chair: ZHOU Yanjie

- 10:50-12:10
1. *Fleet Capacity Delegation for Domestic Cross-Border Transport under Asymmetric Sino-Russian Interests*
CHENG Jiannan (Ningbo University)
 2. *Shore Power Promotion Through Evolutionary Game Theory: Analyzing the Role of National Energy Profiles and Government Policies*
CHEN Dongxu (Ningbo University)
 3. *The Emission Collusion Paradox: How Port Cooperation Under Government Subsidies Undermines Environmental Goals*
JIN Jie (Shanghai Maritime University)
 4. *A Study on the Linguistic Features of IMO Legal Proposals*
ZHU Xiaoling (Shanghai Maritime University)

12:20-14:00

Lunch (3rd Floor Lobby)

Parallel Session 8: Maritime Supply Chain

Chair: Dr. Gao Jinping

- 14:00-15:30
1. *To Share or Not? A Proof From Maritime Supply Chain*
NIAN Xinyue (Shanghai Maritime University)
 2. *Risk-Averse Supplier Selection and Order Allocation Problems Under Supplier Disruption in a Dominant Supplier Market*
Sewon OH (Seoul National University)



3. *Enhancing Supply Chain Resilience in Cruise Shipbuilding: A Fuzzy DEMATEL-ISM-MICMAC Analysis of Capability Structures and Key Drivers*

SHI Yibo (Shanghai Maritime University)

4. *Overseas Industrial Parts Supply Chain Network Design Based on Demand Uncertainty and Inter-Warehouse Transshipment*

ZHAN Beixuan (University of Shanghai for Science & Technology)

5. *Maritime Object Segmentation with Segmentation Anything Model (SAM)*

WU Zhenming (SNF - Center for Applied Research at NHH)

15:30-15:50

Coffee Break (3rd Floor Lobby)

Parallel Session 10: Shipping Network & Digitalization

Chair: Dr. FU Guanghua

1. *Modeling and Performance Evaluation of Shipping Networks under the PortalLite System*

CHU Xin (National University of Singapore)

2. *Advancing Level 5 Digital Twins for Maritime Logistics: Architecture, Infrastructure, and System Integration*

LIU Mengya (National University of Singapore)

3. *Enabling Operational Intelligence in Container Depots through Integrated Digital Twin and Multi-Agent Reinforcement Learning Framework*

LIU Mengya (National University of Singapore)

15:50-17:50

4. *EcoPortMind: A Digital Twin Enabled Framework Integrating Multi-Agent Intelligence for Energy Efficiency Optimization in Container Terminals*

WANG Weihao (Shanghai Maritime University)

5. *Large Language Model-Driven Anomaly Detection in Automated Container Terminal Crane Scheduling*

GU Zixing (Shanghai Maritime University)

6. *A forecast-driven rolling-horizon-based Wagner-Whitin algorithm for establishing a smart coal port inventory system under uncertainties*

DING Peipei (Fuzhou University)

18:30-20:30

Dinner & Banquet (Radisson Collection Lingang Shanghai)



August 28 (Thursday) : Shipping Science Building B304

Parallel Session 5: Automated Terminal Smart Operations

Chair: Dr. GAO Yinping

- 09:00-10:30
1. *Enhanced Yard Template Planning for Container Terminals: A Two-Level POD-Based Approach*
Seung Chan CHOI (Dankook University)
 2. *Spatio-Temporal-Coupled Multi-Resource Collaborative Allocation for Energy-Efficient Automated Container Terminals*
WANG Zhu (Shanghai Maritime University)
 3. *Optimizing AGV Utilization and Battery Life in Automated Container Terminals: Focus on a Novel Charging Strategy and Reinforcement Learning Algorithm*
ZHAO Rui (Shanghai Maritime University)
 4. *An Improved NSGA-II for Joint Scheduling of Handling Equipment in Automated Container Terminals Considering AGV Blocking*
CHEN Xiazhong (Shanghai Maritime University)
 5. *Real-Time Decision Making for Container Stacking Using Machine Learning-Based Weight Clustering*
Seokchan LEE (Dankook University)

10:30-10:50

Coffee Break (3rd Floor Lobby)

Parallel Session 7: Machine Learning Applications

Chair: Prof. CHEN Xinqiang

- 10:50-12:10
1. *A Reinforcement Learning Approach to Solve the Stochastic Container Trans-Marshalling Problem*
QIN Wei (Shanghai Jiao Tong University)
 2. *Inventory Routing Problem of Automotive Parts Considering Time-varying Demands: A Machine Learning Enhanced Branch-and-Price Approach*
ZHENG Renrong (Shanghai Maritime University)
 3. *A Dueling Deep Q-Network and Non-Dominated Sorting-Based Hyper-Heuristic Algorithm for Hybrid-Load AGV Scheduling in Mixed-Flow Assembly Lines*
WANG Zhu (Shanghai Maritime University)
 4. *ALMA: Adaptive Lag Moment Alignment for Domain-Adaptive Time Series Forecasting of Ship Fuel Consumption*
Jung Ho CHOO (Pusan National University)

12:20-14:00

Lunch (3rd Floor Lobby)

Parallel Session 9: Port Energy Supply & Strategies

Chair: Prof. WANG Weili

- 14:00-15:30
1. *Strategic Electricity Sourcing for Green Ports: A Markov Stackelberg Game Approach*
DAI Mengjia (NHH Norwegian School of Economics)



2. *Equipment Selection and Investment Optimization for Alternative Fuel Bunkering Systems in Ports*

LI Xiaoying (Shanghai Maritime University)

3. *Government subsidizes on alternative fuel bunkering facilities at port*

XIE Jiahui (Shanghai Jiao Tong University)

4. *How can ports navigate the emissions reduction requirements set by international policy*

Thalis ZIS (Cyprus University of Technology)

5. *Collaborative Scheduling of Handling Equipment in Automated Container Terminals with Limited AGV-Mates Considering Energy Consumption*

YANG Xurui (Shanghai Maritime University)

15:30-15:50

Coffee Break (3rd Floor Lobby)

Parallel Session 11: Logistics VRP Methods

Chair: Dr. WANG Yu

1. *Robust Optimization on Dry Dock Scheduling Problem*

FAN Lubin (The Hong Kong University of Science and Technology (Guangzhou))

2. *A Branch-and-Price-and-Cut Algorithm for the Collaborative Manned-Unmanned Vehicle Routing Problem with Customer Satisfaction*

LU Houjun (Shanghai Maritime University)

3. *Routing and Delivery with Permission of Delay and Restriction on Drivers' Overtime*

15:50-17:50 Koichi NAKADE (Nagoya Institute of Technology)

4. *Buffer Times in Straddle Carrier Routing*

Ahmet CÜREBAL (University of Hamburg)

5. *Last-Mile Delivery in Community Group Buying: A Contiguity-Aware Approach with Driver-Centric Consideration*

ZHANG Zengni (Tsinghua University)

6. *Managing Truck Service to Mitigate Congestion in a Container Terminal*

Zhuoran AO (The Hong Kong University of Science and Technology (Guangzhou))

18:30-20:30

Dinner & Banquet (Radisson Collection Lingang Shanghai)

August 28 (Thursday) : Shipping Science Building B305

Invited Session: Multi-modal Terminals and Transportation: Data driven approach

Chair: Prof. HE Junliang

10:00-10:30

Invited Talk 10: *Strategic Planning for Containership Green Transition under Net-Zero Emission Target: A Predict-then-Optimize Framework*

Prof. JIN Jiangang, Shanghai Jiao Tong University, China.

10:30-11:00

Invited Talk 11: *Column generation approach for the problem of integrated operations in a container terminal under carbon emission tax*

Prof. WANG Tingsong, Shanghai University, China.



11:00-11:30	Invited Talk 12: <i>The new pattern of global shipping carbon emissions under the influence of the IMO's new decarbonization regulations and the EU emissions trading system</i> Prof. XIAO Guangnian, Shanghai Maritime University, China. Invited Talk 13: <i>AI-Driven Carbon Peaking Optimization for Container Intermodal Transport Networks</i> Prof. HE Junliang, Shanghai Maritime University, China.
12:20-14:00	Lunch (The Third Floor Lobby)
Multi-modal Terminals and Transportation (I) Chair: Prof. YU Hang	
	1. <i>Carbon Reduction and Peaking Pathway Evaluation of Container Intermodal Network: From the Perspective of Policy Sentiment</i> GUO Xiaoyan (Henan University of Technology) 2. <i>Inter-terminal Collaboration: Optimizing Multi-terminal Berth Allocation with Container Ground Transfers</i> 14:00-15:20 ZHANG Di (Shanghai Jiao Tong University) 3. <i>How Emissions Trading System Affects Liner Ship Disruption Recovery</i> LI Shihao (Shanghai University) 4. <i>A Novel Method for Yard Space Allocation for the Port with Atypical Container Collection Cycles</i> LIU Yilin (Shanghai Maritime University)
15:30-15:50	Coffee Break (3rd Floor Lobby)
Multi-modal Terminals and Transportation (II) Chair: Prof. GUO Xiaoyan	
	1. <i>Optimizing Electric Bus Charging Scheduling with PV-ESS Integration for Sustainable Urban Transit</i> XIAO Yu (Shanghai Maritime University) 2. <i>Integrated Optimization of Berth and Quay Crane under Dynamic Multi-vessel Double-berth Mooring Mode</i> 15:50-17:10 GAN Chun (Shanghai Maritime University) 3. <i>Considering the location selection and capacity planning of charging stations for electric trucks with heterogeneous driving ranges</i> TONG Hailin (Shanghai Maritime University) 4. <i>Research on the optimization of energy replenishment for inland electric container ships with integrated charging and swapping</i> WU Hao (Shanghai Maritime University)
18:30-20:30	Dinner & Banquet (Radisson Collection Lingang Shanghai)

Keynote & Invited Presentations

Keynote Presentation I

Ports on their way to the future: examples from Rotterdam

Prof. Rommert DEKKER

Erasmus University, Netherlands

Abstract: Large ports are often accompanied by industrial complexes. Houston, Singapore and Rotterdam / Antwerpen have huge petrochemical industries, as ports are ideal places in the transport chains of many products. Yet changing global trade relations and ongoing drives to sustainability create an unclear future. Ports face congestion on both sea and land side, as well as changes in bunker fuels. We discuss how research brings solutions to these. We discuss ongoing work on alternative fuels, barge and truck congestion, synchronomodality and transport electronic platforms.



Rommert DEKKER is a full professor of Quantitative Logistics and Operations Research at the Erasmus University Rotterdam. He has been doing research in many areas, like maritime, service and reverse logistics. In the maritime area he has been working both on liner shipping network design, terminal automation and hinterland transportation. In the service logistics sector he has cooperated with several companies on optimizing their spare parts logistics. Finally, he was one of the first studying reverse logistics. He recently cooperated with a start-up company on a new concept for e-

commerce returns.



Keynote Presentation II

Optimizing liner shipping routes and speed profiles considering sulfur emission control zones and carbon taxes

Prof. GE Ying-En

Chang'an University, China

Abstract: Sulfur Emission Control Areas (ECA) and carbon tax policies have been progressively implemented worldwide, and the cost to liner shipping companies has accordingly increased. By integrating carbon tax costs, late arrival penalties, fixed costs, and fuel costs, this work proposes a liner shipping route and speed optimization problem to minimize total voyage costs. From an economic perspective, avoidance strategies can effectively reduce total costs in most cases. From an environmental perspective, avoidance strategies reduce sulfur and carbon emissions within the ECA. Extending the time window enables vessels to adopt more fuel-efficient speeds, thereby reducing both delay penalty costs and fuel consumption. Notably, on long-haul routes with greater speed optimization potential, carbon tax hikes yield more pronounced emission reductions.



GE Ying-En, since September 2021, has been Professor and Dean of the School of Transportation Engineering, Chang'an University, Xi'an, China, and serves as Editor-in-Chief for Transportation Research Part D: Transport and Environment, and Associate Editor for Transport Policy. His primary academic interests include transportation network modeling and analysis, transportation and the environment, transportation economics and policy, and operations and management of ports & shipping. He worked in the USA and in the UK for more than 10 years.

Keynote Presentation III

Digital Twins in Seaports and Container Terminals

Prof. Stefan VOSS

University of Hamburg, Germany

Abstract: Digital twins are seen as important drivers of the digitalization in seaports by providing a basis for higher transparency, control and data-driven decision making. We present an exploratory study of digital twins in seaports based on a literature review and case studies, especially focusing on a case from Hamburg, Germany. The analysis reveals a standardization deficit for digital twin implementations, an inflationary and improper use of the term digital twin, and fields of research that need to be explored further. An in-depth examination of multiple case studies and discussions with global port leaders yields valuable perspectives on the varied levels of digital twin implementations being applied today. As a result of the analyses conducted, various research directions and a research agenda are presented.



Stefan VOSS is professor and director of the Institute of Information Systems at the University of Hamburg. His current research interests are in quantitative / information systems approaches to supply chain management and logistics including public mass transit and telecommunications. He is author and co-author of several books and several hundred papers in various journals. In the German Handelsblatt ranking he is continuously within the top 20 professors in business administration within the German speaking countries. The most recent Wirtschaftswoche rankings list him among the top 10. He serves on the editorial board of some journals including being Editor of Public Transport.

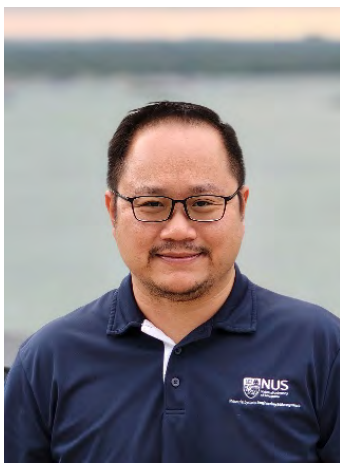
Keynote Presentation IV

Beyond the Data: Rethinking AI Applications for Logistics and Maritime Systems

Prof. Haobin LI

National University of Singapore, Singapore

Abstract: In recent years, data-driven AI has achieved remarkable success in consumer markets. However, when we shift our focus to logistics and maritime systems, this paradigm does not always hold. In such contexts, rule understanding becomes critical. Excellence in applying AI to logistics and maritime systems lies in the ability to differentiate between these two domains: when to leverage simulation-based, rule-oriented AI approaches such as generative or agentic AI, and when to rely on proven data-driven methods. This keynote will explore the interplay between these paradigms, illustrate the risks of misapplication, and propose a framework for strategically combining them to unlock AI's full potential in an increasingly complex and uncertain maritime logistics environment.



Haobin LI is a Senior Lecturer in the Department of Industrial Systems Engineering and Management at the National University of Singapore, Co-Director of the Centre of Excellence in Modelling and Simulation for Next Generation Ports, and Academic Director of the MSc in Maritime Technology and Management programme. His research specialises in discrete-event simulation and stochastic simulation optimisation for complex industrial systems, with applications in next-generation logistics, maritime ports, and smart manufacturing. He is the developer of O2DES.NET, an object-oriented simulation and optimisation framework that underpins C4NGP's core technology and several major national port projects.

Keynote Presentation V

Navigating the Future of Shipping: AI Innovations and Their Impact

Prof. Xiuju FU

*Institute of High Performance Computing, Agency for Science, Technology and Research
(A*STAR), Singapore.*

Abstract: This talk will provide a comprehensive overview of the Maritime AI Research Programme in Singapore, showcasing key research findings and achievements that demonstrate AI's transformative potential in maritime shipping. From enhancing traffic safety and driving maritime decarbonization to optimizing operational efficiency, the session will explore how AI is reshaping the industry. Additionally, the talk will introduce the System for Maritime AI Excellence (MAX) framework, a strategic initiative aimed at unifying, scaling, and elevating maritime AI efforts across the ecosystem.



Xiuju FU is Senior Principal Scientist and Director of Maritime AI Research Programme at Institute of High Performance Computing, Agency for Science, Technology and Research (A*STAR) in Singapore. With expertise in AI, big data intelligence, simulation, and optimization techniques, she focuses on advancing complex system management and enhancement. Recognized for her contributions, she was honored as an SMI Fellow in 2023. Currently, she spearheads the Maritime AI Research in IHPC, driving research and development initiatives in maritime data excellence, AI modeling excellence, maritime AI computing, and application excellence. Her efforts aim to foster the development and application of big data and AI in the maritime industry.

Keynote Presentation VI

AI-Powered Smart Port Logistics: Lessons from South Korea's National Projects

Prof. Hyerim BAE

Pusan National University, Korea

Abstract: Since ports need to be optimized alongside both maritime and inland logistics, it is crucial to share and utilize data from these diverse systems to enhance the overall flow of goods. Additionally, safety and environmental concerns have become vital factors that must be integrated into optimization strategies within the global supply chain. Advances in Artificial Intelligence (AI) are also driving the adoption of entirely new approaches that go beyond traditional methods. In this keynote, will share three large-scale, nation-level projects conducted in South Korea as case studies to showcase methodologies for developing smart logistics in maritime ports within the global supply chain. Additionally, I will explore how emerging technologies such as generative AI and large language models (LLMs) can be utilized in maritime and port operations.



Hyerim BAE is currently Dean of the Graduate School of Data Science at Pusan National University (PNU) in South Korea and a Professor in the Department of Industrial Engineering at the same university. He also serves as Director of the Human-Centered Carbon Neutral Global Supply Chain Research Center, a multi-year initiative supported by the National Research Foundation of Korea. His work explores the convergence of artificial intelligence, big data analytics, and sustainable logistics, with a particular focus on smart ports and digital supply chains. His research contributions range from process analytics and time-series prediction models to port simulation systems and collaborative learning frameworks for maritime environments.

Keynote Presentation VII

Recent Advances in Multiobjective Sustainable Closed-loop SCM Network Models

Prof. Mitsuo GEN

Tokyo Univ. of Science, Japan

Abstract: The continuous growth of the globalizing economy has led to the prevalence of Multiobjective Sustainable Closed-loop Supply Chain Management (Mo-SCLSCM) network designs. In this talk, we will firstly introduce the logistics flow of forward and reverse in which the forward logistics consisting of Supplier-Manufacturer-Distribution Center-Customer and the reverse logistics consisting of Returning Centers-Reprocessing Centers-Manufacturer are composing a Closed-loop SCM network model. We will summarize the real-world multiobjective reverse logistics models by a hybrid GA based on priAWGA (priority-based encoding by Adaptive Weight GA). We will also introduce recent advances of multiobjective SCLSCM models such as Mo-Sustainable SCM model and various Mo-Sustainable CLSCM model with their effectiveness.



Mitsuo GEN received his PhD in Engineering from Kogakuin University, PhD degree in Informatics from Kyoto University and is now Visiting Prof. at TUS. He was faculties at Ashikaga Institute of Technology for 1974-2003, Waseda University for 2003-2010, and Senior Researcher at Fuzzy Logic Systems Inst. for 2009-2024. He was visiting faculties at Univ. of California at Berkeley for 1999.8-2000.3, Texas A&M Univ. for 2000.1-3 & 2000.8-9, Hanyang Univ. in S. Korea for 2010-2012 and National Tsing Hua Univ. in Taiwan for 2012-2014. His research field is Evolutionary Computation, Manufacturing Scheduling and Logistics.

Invited Talk 1

Green Port Construction and Prospects

FANG Huaijin

Shanghai International Port (Group) Co., Ltd. (SIPG), China

Abstract: Green-port development has become the prime mover of high-quality growth in the nation's waterborne-transport sector. These facilities weave green methanol, LNG into their energy architecture and underpin them with shore power and battery swap/charge infrastructure, closing the loop on zero-emission cargo-handling and vessel operations. This address distils the transformation into three sets of evidence: the six ports as national benchmarks; nine green-shipping corridors launched since 2022 between Chinese; and the Shanghai International Shipping Centre itself. Looking ahead to the 2030 global shipping-emission milestone, Shanghai Port advances the concept of a "Green, Smart, Zero-Carbon Island," anchored on large-scale renewables, an intelligent control ecosystem, and city-port symbiosis, to create a globally replicable port paradigm and deliver a China-crafted blueprint for worldwide port decarbonisation.



FANG Huaijin, serves as Deputy Director of both the National Port Standardization Technical Committee and the National Container Standardization Technical Committee. Formerly CFO and subsequently Vice President of Shanghai International Port (Group) Co., Ltd. (SIPG). Since 2013, he has directed SIPG's capital projects and technological innovation of the world's largest fully automated container terminal—Yangshan Phase IV. Under his leadership, the terminal's intelligent, green intermodal system achieved large-scale deployment, establishing a flagship brand, Chinese manufacturing, Chinese services, and Chinese standards for smart, green ports.

Invited Talk 2

From Practice to Foresight: Smart Innovation in Intelligent Logistics and the Digital Development Path of Shipping & Trade

WANG Beili

Shanghai E&P International, Inc, China

Abstract: Shanghai E&P International, Inc. is the core builder and operator of Shanghai's large-scale port logistics information system and a backbone enterprise in the development of Shanghai International Shipping Center and International Trade Center. As the primary maintenance and service provider for the Shanghai International Trade Single Window, the company will share real-world cases illustrating how it enables the construction of an intelligent logistics system and will introduce a next-generation shipping-and-trade digital platform. Driven by a dual-engine approach—technical innovation and model innovation—this platform will create a full-chain digital system of “data connectivity, business synergy, and intelligent decision-making,” reshaping process efficiency across the shipping and trade industry and injecting intelligent momentum into high-quality development.



WANG Beili is Deputy Director of the Business Development Center at Shanghai E&P International, Inc.. She has led teams in the deep-level delivery of multiple landmark municipal projects, including the Shanghai Port “One-Stop” Customs Clearance initiative, the Shanghai E-Port platform, and the Shanghai International Shipping Center platform. Leveraging her extensive hands-on experience and outstanding management

capabilities, she continues to strengthen the development of Shanghai as an international shipping and trade hub, playing a vital role in improving China's cross-border trade environment.

Invited Talk 3

Practical Issues and Reflections on the Development of Multimodal Transport in the Yangtze River Economic Belt

Prof. YANG Bin

Shanghai Maritime University, China

Abstract: This presentation focuses on the current practical challenges facing multimodal transport development in the Yangtze River Economic Belt. Topics include optimizing multimodal solutions for new-energy vehicle exports, standardizing battery-swap systems for electric vessels, aligning cargo categories and standards between rail and maritime transport, integrating resources along long-haul alliance routes, analyzing Shanghai's freight volumes and transport structure, optimizing comprehensive transportation tolls, and establishing a maritime arbitration alliance. The talk will also highlight intriguing academic research questions inspired by these issues.



YANG Bin serves as a doctoral supervisor in both Management Science and Engineering and Transportation Engineering, as well as Dean of the Graduate School and Chair of the Degree Evaluation Committee for Logistics Management and Engineering. He concurrently holds positions such as Head of the Logistics Management Group under the Logistics Education Steering Committee of China's Ministry of Education and Vice President of the China Society of Logistics. He has supervised more than 100 master's students and 7 doctoral candidates in Management Science and Engineering and Transportation Engineering. He has authored over 100 academic papers and has completed more than 100 research projects funded by the National Natural Science Foundation of China and other agencies.

Invited Talk 4

Navigating the High Seas: Reshaping Global Trade and Supply Chains in a US Tariff-Driven World

Prof. Paul Tae-Woo LEE

Zhejiang University, China

Abstract: This presentation aims to briefly address the impact of US tariff policy and the reshoring or establishment of production lines on maritime transport and logistics from the perspective of stakeholders, such as shipping companies, port authorities, terminal operators, and freight forwarders. It also illuminates China's counter-strategy of establishing global logistics distribution centers along the Belt and Road, which may mitigate tariff impacts by enhancing the resilience of the trade and maritime transport and logistics with the help of the 6th-generation port model as smart ports and green and digital shipping corridors. By navigating these "high seas," this speech may contribute to inspiring industry leaders to build efficient, sustainable maritime transport and logistics systems in a US tariff-driven world from the Chinese perspective.



Paul Tae-Woo LEE is PhD Supervisory Professor of Maritime Transport and Logistics and Director of the Maritime Logistics and Free Trade Islands Research Centre at Ocean College, Zhejiang University. He is currently an Adjunct Professor at RMIT University, Australia. He is a recognized speaker at international forums such as APEC, UNDP, UNESCAP, China Academy of Social Sciences, and Vietnam Academy for Social Sciences, and currently serves as a consultant to UNESCAP, producing influential reports on maritime connectivity, green shipping corridors, sustainable port development, and smart ports.

Invited Talk 5

AI-supported Human Factors in ATM and VTM Research

Prof. Chun-Hsien CHEN

Nanyang Technological University, Singapore

Abstract: Artificial intelligence (AI) technologies have been applied to support various human factors tasks in recent years. This speech will cover some aspects of AI in human factors, inclusive of background: AI and human factors, the role of AI in human factors, applications of AI in human factors for air traffic management (ATM), and applications of AI in human factors for vessel traffic management (VTM).



Chun-Hsien CHEN is Full Professor, Director of the Design Stream, and Professor-in-Charge of the Design & Human Factors Lab in the School of Mechanical & Aerospace Engineering, Nanyang Technological University, Singapore. He received his BS degree in Industrial Design from National Cheng Kung University, Taiwan, MS and Ph.D. degrees in Industrial Engineering from the University of Missouri-Columbia, USA. He has several years of product design & development experience in the industry. His teaching and research interests are design science in product design and development; engineering/design informatics for managing / supporting digital design & manufacturing; and human factors and management of human performance. He has more than 300 publications in these areas.

Invited Talk 6

Development of green maritime logistics industry: Coordinated policy making and interdisciplinary studies

Prof. FENG Xuehao

Zhejiang University, China

Abstract: The size of maritime logistics is increasing with the development of global trade and supply chain. However, the carbon neutrality in the maritime logistics industry also requires more efforts. In this talk, we would like to discuss some recent practice in Chinese and European ports who implemented green technologies and facilities. The related policy making to support these implementations will be discussed. Second, we would like to talk about how decentralized policy makings might fail to achieve the expected green target. Our third part is about the integration of multi-modal transportation network based on the digital supply chain practice. We hope to highlight the importance of interdisciplinary studies and talent training in this area for the sustainable development of the industry.



FENG Xuehao is a full professor of State Key Laboratory of Ocean Sensing and Ocean College at Zhejiang University, China. His research interests lie in the areas of maritime logistics, green port and shipping, and marine economy. He has published 40 papers on SSCI/SCI journals, many of which are top-tier such as Transportation Research Part B, Transportation Research part A, Omega, Maritime Policy & Management. He served as the associated editor of SSCI/SCI journals, Maritime Policy & Management and European Journal of Industrial Engineering.

He also conducted several national research projects as PI in the field of maritime logistics and supply chain management.

Invited Talk 7

An integrated framework for truck arrival management and yard capacity utilization in a container terminal

Prof. JIA Shuai

The Hong Kong University of Science and Technology, China

Abstract: Truck congestion in a container terminal yard is usually caused by the imbalance between the arrival demand of external trucks and the supply of yard service capacity. This paper proposes an integrated modelling framework that mitigates terminal yard congestion by joint optimization of the truck arrival demand and the utilization of yard capacity, subject to truck arrival demand shifting, yard queueing, and yard resource allocation constraints. For solving the integrated model, we develop a tailored iterative solution method. We evaluate the computation performance and solution quality of the integrated framework on instances generated from the operational data of a large container terminal. Based on the computation results, we provide insights into truck demand and service management for congestion mitigation in a container terminal.



JIA Shuai is Assistant Professor of Thrust of Intelligent Transportation, The Hong Kong University of Science and Technology (Guangzhou). His research areas include operations research, artificial intelligence, and simulation, with applications in multimodal transportation, logistics and supply chain, and resource allocation and scheduling. He has published over 30 papers in prestigious academic journals such as Transportation Science and Transportation Research Parts B/C/D/E, and has

secured research grants as a PI from various funding sources.

Invited Talk 8

Lecture on artificial intelligence and digital port logistics

Prof. DING Yi

Shanghai Maritime University, China

Abstract: This lecture took the port logistics platform system as an example, combined with the intelligent and digitalized port logistics, to analyze the specific application ways of AI methods such as machine learning and operational research optimization in the complex problems of port logistics. And the case study in the lecture is of great benefit to learning the in-depth application of blockchain, big data, AI intelligent algorithm, VR technology and digital twin technology in port logistics. In the meanwhile, the construction of integrated port logistics management and control platform based on digital twin technology are summarized. This lecture is helpful to understand the digital development of the specific business of various entities.



DING Yi, PhD, Professor and doctoral supervisor of Logistics Research Center, Shanghai Maritime University, Founder and General Manager of Milo Information Technology Co. Ltd. He has been committed to the research direction of port operation optimization and information system, and has published more than fifty academic papers in this field. As an expert in the field of port logistics industry informatization, he has more than fifteen years' experience in large-scale port logistics information system research and technical team management. The multiple AI and big

data systems he developed, such as automatic stowage, intelligent berth, port big data and terminal machine vision, have been successfully applied in port logistics enterprises in Shanghai, Dalian, Xiamen, Anhui and other places.

Invited Talk 9

Innovative Models in Port Collection-Distribution Systems — Underground Logistics Applications

Prof. LIANG Chengji

Shanghai Maritime University, China

Abstract: With the vigorous development of global trade and the deepening implementation of the "dual carbon" strategy, port collection and distribution systems face dual pressures of efficiency improvement and low-carbon transformation. Traditional road-dominated transport models have led to increasingly prominent issues such as traffic congestion, excessive carbon emissions, and port-city conflicts. As an innovative transport solution, the Underground Container Logistics system has emerged as a crucial pathway to address these challenges, leveraging its high efficiency, low-carbon, and independence from surface traffic. This lecture focuses on discussing and analyzing emerging collection and distribution models for container ports, while introducing the development of underground logistics systems and their application in port operations.



LIANG Chengji is Professor and PhD Supervisor at Shanghai Maritime University, and Chair of the Academic Committee of the Institute of Logistics Science and Engineering. Her primary research focuses on port logistics, specifically the operation planning and optimization of port logistics systems, resource allocation/dispatch optimization and simulation, and port layout optimization and simulation. Recent work explores automated container terminals, underground container logistics systems, and green port energy transformation. Currently, she holds the positions of Vice Chair of the Underground Logistics Committee and Council Member of the Underground Space Division within the Chinese Society for Rock Mechanics & Engineering.

Invited Talk 10

Strategic Planning for Containership Green Transition under Net-Zero Emission Target: A Predict-then-Optimize Framework

Prof. JIN Jiangang

Shanghai Jiao Tong University, China

Abstract: Shipping companies are under growing pressure to meet net-zero emission targets under tightening regulations. To tackle this challenge, this paper proposes a predict-then-optimize framework for designing green transition strategies. The framework integrates a vessel-specific fuel consumption prediction model for comprehensive monitoring of emissions in future years, and an optimization model that strategically schedules and combines multiple emission reduction methods. The model minimizes total lifecycle costs, while ensuring compliance with carbon intensity indicator regulations and the overall net-zero emission target. A case study on several container ships demonstrates the applicability of the model, with numerical experiments showing that the predict-then-optimize framework achieves a 67% carbon reduction by 2040. Further investigation of the strategies is conducted to provide insights into optimal strategy design.



JIN Jiangang is a Professor of the School of Ocean & Civil Engineering at Shanghai Jiao Tong University. His research interests lie in the application of large-scale modeling and optimization to transportation and logistics systems. He received the National Excellent Young Scholars Award of National Natural Science Foundation of China, the First Price of INFORMS RAS Problem Solving Competition. He has published more than 50 referred papers in international journals and conferences, such as Transportation Science, Transportation Research Part B/C/E. He is serving as Area Editor for the journal of Computers & Industrial Engineering.

Invited Talk 11

Column generation approach for the problem of integrated operations in a container terminal under carbon emission tax

Prof. WANG Tingsong

Shanghai University, China

Abstract: The International Maritime Organization has proposed to impose carbon emission tax on ports to reduce their carbon emission and sustain the green development of maritime industry. The berth allocation, QC/YC assignment, truck deployment and subblock reservation are the operations for vessels to moor and stevedore containers, and they mutually interact with each other, which consequently affect the stevedoring efficiency and the carbon emission. Therefore, this paper tries to integrate operations with QC maintenance under carbon emission tax. To solve the model on large-scale instances, a column generation (CG) based approach is developed. The results indicate that the carbon taxation policy has significant impact on integrated operation plans, and can significantly reduce the carbon emission. Also, we compare the effects of stepwise and unitary carbon taxation policies on carbon emission reduction, and find the former performs better than the latter. In addition, computation results show that the proposed CG based approach has better performance compared with commercial solvers.



WANG Tingsong is currently a Professor in the School of Management, Shanghai University, and he is the Director of the Department of Management Science and Engineering. He is active academically and professionally in the area of operations research and management science, particularly in the area of maritime logistics and management, network modeling and optimization, risk control and analysis. He has published more than 50 technical papers in leading international journals and conferences.

Invited Talk 12

The new pattern of global shipping carbon emissions under the influence of the IMO's new decarbonization regulations and the EU emissions trading system

Prof. XIAO Guangnian

Shanghai Maritime University, China

Abstract: With the expansion of maritime trade, the issue of carbon emissions caused by maritime transportation is receiving increasing attention. Under the combined influence, shipping routes are being re-arranged. The cost structure of the shipping industry has also been significantly altered. Shipowners and operators face increasing costs for emissions, compelling them to invest in low-carbon technologies. This new pattern not only poses challenges to the traditional operation model of the shipping industry but also brings opportunities for the development of green shipping technologies. It calls for coordinated efforts from the international community, shipping companies, and relevant stakeholders to balance emission reduction goals with the sustainable development of the shipping industry.



XIAO Guangnian is Vice Dean of the School of Economics and Management at Shanghai Maritime University. His primary research focuses on intelligent transportation and green shipping. He has led two National Natural Science Foundation projects, a Shanghai Youth Soft Science Research Project, a research project for the Decision-Making Consultation of the Shanghai Municipal Government. He has published over 40 papers in international journals in the fields of transportation and shipping and currently serves as a Youth Editorial Board Member for the journal

Sustainable Horizons.

Invited Talk 13

AI-Driven Carbon Peaking Optimization for Container Intermodal Transport Networks

Prof. HE Junliang

Shanghai Maritime University, China

Abstract: The global consensus on carbon peak and carbon neutral goals extends to the transport sector. An approach for predicting the carbon emission peak time and peak value within a container intermodal network (CIN) is proposed. Our approach commences by capturing the fundamental network elements, through intermodal network modeling. We then integrate a carbon emission model with scenario analysis, utilizing Monte Carlo simulation to predict the probability of the CIN's carbon emission peak time and value. Finally, using the Yangtze River Delta regional container intermodal network (YRD-CIN) as a case study, 15 scenarios are set up to explore the effects of intermodal transport network development on carbon peaks and peak times. The findings demonstrate the significant influence of modal transport shifts, network connectivity, and energy technology levels on the carbon peak in the YRD-CIN.



HE Junliang is currently the deputy director of the Institute of Logistics Science and Engineering, Shanghai Maritime University. He was awarded the Shanghai High-level Talent Program, Shanghai Shuguang Scholars, Shanghai Talent Development Funding Program, Shanghai Chenguang Scholars, Shanghai Yangfan Program, Pudong New District Mingzhu talent Program, and other provincial and ministerial talent programs. He is an editorial board member of a SCI journal, and editor-in-chief of the special issues of SCI international journals AEI and ANOR.



Paper Presentations

HU Hongtao, Shanghai Maritime University

An Investigation into the Optimization of Vessel Unloading Scheduling for Triple Hoist Quay Cranes in Two-Story Container Terminals

WANG Zhu, Shanghai Maritime University

Berth and Quay Cranes Allocation Problem with On-Shore Power Supply Assignment in Container Terminals

YANG Xurui, Shanghai Maritime University

Collaborative Scheduling of Handling Equipment in Automated Container Terminals with Limited AGV-Mates Considering Energy Consumption

Sang Duck LEE, Korea Railroad Research Institute

Learning-based Assessment for Port Occupational Safety

JIA Haiying, Norwegian School of Economics

What Weather is Good Weather under Performance Clauses in Charterparties?

CHENG Jiannan, Ningbo University

Fleet Capacity Delegation for Domestic Cross-Border Transport under Asymmetric Sino-Russian Interests

Seokchan LEE, Dankook University

Real-Time Decision Making for Container Stacking Using Machine Learning-Based Weight Clustering

DING Peipei, Fuzhou University

A data-driven rolling-forecast-horizon LSTM-based Wagner-Whitin algorithm for establishing a smart coal port inventory system under uncertainties

DAI Mengjia, NHH Norwegian School of Economics

Strategic Electricity Sourcing for Green Ports: A Markov Stackelberg Game Approach

QIN Wei, Shanghai Jiao Tong University

A Reinforcement Learning Approach to Solve the Stochastic Container Trans-Marshalling Problem

ZHANG Di, Shanghai Jiao Tong University

Inter-terminal Collaboration: Optimizing Multi-terminal Berth Allocation with Container Ground Transfers

DU Juan, Shanghai Maritime University

An Analogical Corpus-Based Study of Maritime Proposal Convergence Trends for Chinese and English Native Speakers

Kyunghun LEE, Pusan National University

Multimodal Sensor Projection and Feature-Based Clustering for Abnormal Vessel Detection



ZHU Xiaoling, Shanghai Maritime University

A Study on the Linguistic Features of IMO Legal Proposals

CHEN Dongxu, Ningbo University

Shore Power Promotion Through Evolutionary Game Theory: Analyzing the Role of National Energy Profiles and Government Policies

Seung Chan CHOI, Dankook University

Enhanced Yard Template Planning for Container Terminals: A Two-Level POD-Based Approach

Reinier VAN UDEN, Erasmus University Rotterdam

A Rolling-Horizon Model for Fair and Efficient Just-In-Time Arrival of Ships

Seongbae JO, Seoul National University

Stack-Based Yard Allocation Considering Inter-Terminal Transportation in a Transshipment Hub

Sewon OH, Seoul National University

Risk-Averse Supplier Selection and Order Allocation Problems Under Supplier Disruption in a Dominant Supplier Market

Minsu WOO, Dankook University

Cost-Efficient Quay Crane Assignment and Scheduling Under Time-Variant Surcharge Policies in Container Terminals

Thalis ZIS, Cyprus University of Technology

How Can Ports Navigate the Emissions Reduction Requirements Set by International Policy

LIU Yilin, Shanghai Maritime University

A Novel Method for Yard Space Allocation for the Port with Atypical Container Collection Cycles

HE Junliang, Shanghai Maritime University

Integrated Optimization of Berth and Quay Crane under Dynamic Multi-vessel Double-berth Mooring Mode

LI Shihao, Shanghai University

How Emissions Trading System Affects Liner Ship Disruption Recovery

WU Hao, Shanghai Maritime University

Research on the Optimization of Energy Replenishment for Inland Electric Container Ships with Integrated Charging and Swapping

ZHAO Rui, Shanghai Maritime University

Optimizing AGV Utilization and Battery Life in Automated Container Terminals: Focus on a Novel Charging Strategy and Reinforcement Learning Algorithm

GUO Xiaoyan, Henan University of Technology

Carbon Reduction and Peaking Pathway Evaluation of Container Intermodal Network: From the Perspective of Policy Sentiment



LU Houjun, Shanghai Maritime University

A Branch-and-Price-and-Cut Algorithm for the Collaborative Manned-Unmanned Vehicle Routing Problem with Customer Satisfaction

LI Xiaoying, Shanghai Maritime University

Equipment Selection and Investment Optimization for Alternative Fuel Bunkering Systems in Ports

CHU Xin, National University of Singapore

Modeling and Performance Evaluation of Shipping Networks under the PortalLite System

LIU Mengya, National University of Singapore

Enabling Operational Intelligence in Container Depots through Integrated Digital Twin and Multi-Agent Reinforcement Learning Framework

LIU Mengya, National University of Singapore

Advancing Level 5 Digital Twins for Maritime Logistics: Architecture, Infrastructure, and System Integration

SHAO Changying, National University of Singapore

Activity-based or Behaviour-based? A Comparative Evaluation of Hybrid Modeling Formalism for Container Terminal Operations

WU Zhenming, SNF - Center for Applied Research at NHH

Maritime Object Segmentation with Segmentation Anything Model (SAM)

WEI Hairui, University of Shanghai for Science & Technology

Study on Coordination Mechanism and Pricing of Lithium Carbonate Option Contract in Closed-Loop Power Battery Supply Chain under Supply Constraints in Recycling Market

ZHAN Beixuan, University of Shanghai for Science & Technology

Overseas Industrial Parts Supply Chain Network Design Based on Demand Uncertainty and Inter-Warehouse Transshipment

QU Weijia, Shanghai Maritime University

Mixed Car-Carrier Fleets Capacity Design for Finished-Vehicle Logistics: A Column Generation Method

TANG Dong, Shanghai Maritime University

Reinforcement Learning-Based Dynamic Slot Allocation In Container Yards Considering Container Cluster Strategy

CHEN Shiqi, Shanghai Maritime University

Carbon Emission Reduction Potential of Express Delivery Transportation under Road-to-Rail Policy

GAO Yinping, Shanghai Maritime University

Optimization of Multi-Energy Rubber-Tired Crane Scheduling under Uncertainty

Koichi NAKADE, Nagoya Institute of Technology

Routing and Delivery with Permission of Delay and Restriction on Drivers' Overtime



WANG Xudong, Shanghai Maritime University

Joint Vessel Deployment Strategy Considering Individual Operational Preferences

JIN Jie, Shanghai Maritime University

The Emission Collusion Paradox: How Port Cooperation Under Government Subsidies Undermines Environmental Goals

ZHAO Runqing, The Hong Kong University of Science and Technology (Guangzhou)

Robust Vessel Traffic Scheduling with Uncertain Berth Service Times in a Seaport

DING Peipei, Fuzhou University

A Forecast-Driven Rolling-Horizon-Based Wagner-Whitin Algorithm for Establishing a Smart Coal Port Inventory System Under Uncertainties

AO Zhuoran, The Hong Kong University of Science and Technology (Guangzhou)

Managing Truck Service to Mitigate Congestion in a Container Terminal

WANG Zhu, Shanghai Maritime University

A Dueling Deep Q-Network and Non-Dominated Sorting-Based Hyper-Heuristic Algorithm for Hybrid-Load AGV Scheduling in Mixed-Flow Assembly Lines

NIAN Xinyue, Shanghai Maritime University

To Share or Not? A Proof From Maritime Supply Chain

FAN Lubin, The Hong Kong University of Science and Technology (Guangzhou)

Robust Optimization on Dry Dock Scheduling Problem

WANG Zhu, Shanghai Maritime University

Spatio-Temporal-Coupled Multi-Resource Collaborative Allocation for Energy-Efficient Automated Container Terminals

XIE Jiahui, Shanghai Jiao Tong University

Government Subsidizes on Alternative Fuel Bunkering Facilities at Port

Jung Ho CHOO, Pusan National University

ALMA: Adaptive Lag Moment Alignment for Domain-Adaptive Time Series Forecasting of Ship Fuel Consumption

LIU Yun, Shanghai Jiao Tong University

Drone Scheduling Optimization for Continuous Sea Area Emission Monitoring

ZHANG Zengni, Tsinghua University

Last-Mile Delivery in Community Group Buying: A Contiguity-Aware Approach with Driver-Centric Consideration

SHI Yibo, Shanghai Maritime University

Enhancing Supply Chain Resilience in Cruise Shipbuilding: A Fuzzy DEMATEL-ISM-MICMAC Analysis of Capability Structures and Key Drivers

MA Zhongjun, Tsinghua University

How Reservation and Customer Ranking Mechanism Affect Efficiency, Revenue and Fairness in the Panama Canal's Transits?



GUO Yanru, Guangzhou Maritime University

A Dynamic Stochastic Optimization Framework for De-risking Green Port Investments: The Case of Shore Power in Guangzhou

DONG Xinxin, Shanghai Maritime University

Intelligent Positioning Method for Ship Block Yard Based on Spatial-Temporal Bi-Objective Constraints

WANG Weihao, Shanghai Maritime University

EcoPortMind: A Digital Twin Enabled Framework Integrating Multi-Agent Intelligence for Energy Efficiency Optimization in Container Terminals

Ahmet CÜREBAL, University of Hamburg

Buffer Times in Straddle Carrier Routing

WANG Yu, Shanghai Maritime University

Inventory Routing Problem of Automotive Parts Considering Time-varying Demands: A Machine Learning Enhanced Branch-and-Price Approach

YU Mingzhu, Shenzhen University

Ocean Container Shipping Pricing Considering Empty Container Repositioning and Containerized Bulk Cargo Transportation

CHEN Xiazhong, Shanghai Maritime University

An Improved NSGA-II for Joint Scheduling of Handling Equipment in Automated Container Terminals Considering AGV Blocking

GU Zixing, Shanghai Maritime University

Large Language Model-Driven Anomaly Detection in Automated Container Terminal Crane Scheduling

YU Xiao, Shanghai Maritime University

Optimizing Electric Bus Charging Scheduling with PV-ESS Integration for Sustainable Urban Transit

GAN Chun, Shanghai Maritime University

Considering the location selection and capacity planning of charging stations for electric trucks with heterogeneous driving ranges



Industry Tour

- Date** 29 August (Fri.) from 09:00 to 12:00
- Venue** Yangshan Automated Terminal
- Address** Yangshan Phase IV, S2 Hulu Expressway, Yangshan Town, Zhoushan, China
- Directions** Bus departs from Shanghai Maritime University at 9:00 am
- >> Tour Yangshan Automated Terminal
 - >> Return to Shanghai Maritime University at 12:00 pm



Special Issues

Outstanding full papers from the conference will be recommended to the following journals:

- Transportation Research Part D (Special issue in preparation)
- Computer in Industry (Special issue in preparation)
- Maritime Policy & Management (Special issue in preparation)
- Computers & Industrial Engineering
- Engineering Optimization
- European Journal of Industrial Engineering
- Flexible Services and Manufacturing Journal
- Frontiers in Industrial Engineering

Practical Information

Accommodation:

The Organizing Committee has negotiated special rates with the following hotels. Please make your reservations through the following booking channels. Shuttle services between the hotels and the venue will be arranged during the conference.

1. Radisson Collection Lingang Shanghai

- Address: No. 100 Dingke Road, Pudong, Shanghai, China, 201306
- Rates: RMB 500 (single breakfast) / RMB 550 (double breakfast)
- Reservation:
 - Email: reservation.jjrc@thejinjianghotels.com
 - Phone : +86 (021)38819999 transfer to the Reservation Department
 - Mobile: +86 19602170026 (No answer after 8:30 PM on weekdays and after 5:30 PM on weekends)
- Note:
 - Participants should mention **2025 International Conference on Maritime Logistics Systems** or **LOGMS2025** when making internal reservations.
 - Please provide the check-in date, check-out date, guest name, guest's mobile number, and room type when booking.
 - A **guarantee** for the first night's room charge is required.

2. Vienna Int'l Hotel Shanghai Harbour City

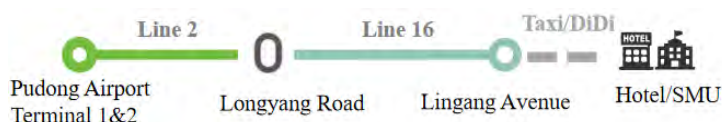
- Address: 22 Kang Nai Xin Road, Pudong, Shanghai, China, 201306
- Rates: RMB 350 (single breakfast) / RMB 398 (double breakfast)
- Reservation: +86 13816346530 (Mobile / Wechat) Manager Xu
- Note: Participants should mention **2025 International Conference on Maritime Logistics Systems** or **LOGMS2025** when making internal reservations.

Transportation:

1. Airport/Railway Station to SMU/Hotel

- Shanghai Pudong International Airport:**

- 1) Taxi/DiDi: ~33 km to SMU, ~34 min (approx. CNY 120)
- 2) Metro + Taxi/DiDi: 1h 55min (CNY 11 for metro + approx. CNY 20 for taxi)



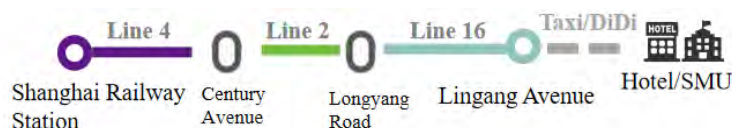
- Shanghai Hongqiao International Airport/ Shanghai Hongqiao Railway Station:**

- 1) Taxi/DiDi: ~87 km to SMU, ~90 min (approx. 290 CNY)
- 2) Metro + Taxi/DiDi: 1h 58min (CNY 11 for metro + approx. CNY 20 for taxi)



- Shanghai Railway Station:**

- 1) Taxi/DiDi: ~77 km to SMU, ~76 min (approx. 280 CNY)
- 2) Metro + Taxi/DiDi: 1h 45min (CNY 10 for metro + approx. CNY 20 for taxi)



- Shanghai South Railway Station:**

- 1) Taxi/DiDi: ~71 km to SMU, ~73 min (approx. 250 CNY)
- 2) Metro + Taxi/DiDi: 1h 49min (CNY 10 for metro + approx. CNY 20 for taxi)



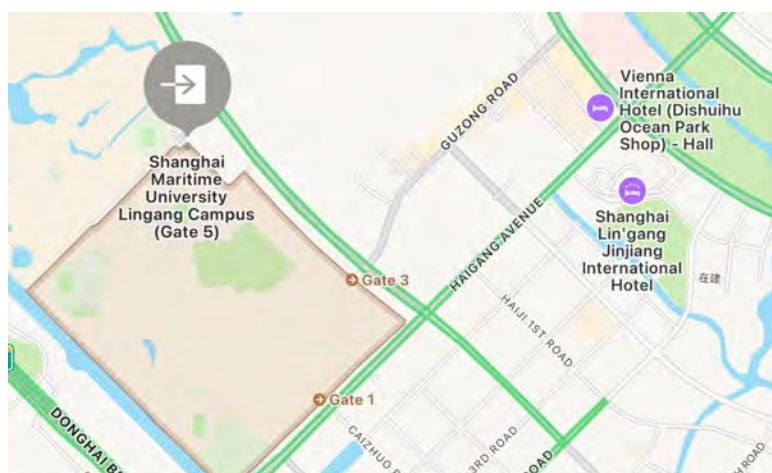


2. Local Transportation Tips

- Taxis and DiDi apps are convenient in Shanghai.
- Avoid peak hours (7:30–9:30 AM, 5:00–7:00 PM) and allow extra travel time.

3. Hotel to SMU

- Shuttle buses will be arranged between hotels and SMU on **August 27 and 28**.
- Departure times will be provided at check-in and adjusted according to the conference schedule.



4. SMU Map

Conference venue: Shipping Science Building (Building 39).





上海海事大学
SHANGHAI MARITIME UNIVERSITY



If you need to park on campus, please complete the registration in advance by scanning the QR code.



活动专用停车码

LOGMS2025国际会议

开始时间: 2025-08-26 00:00:00
结束时间: 2025-08-29 23:59:59

温馨提示:

- 1.扫描录入车牌;
- 2.此二维码仅限于本场活动使用;
- 3.最终解释权归上海海事大学保卫处所有;

We are looking forward to meeting you in Shanghai, China!

Please follow the official website for updates. (<https://logms2025.shmtu.edu.cn/main.htm>)



LOGMS@shmtu.edu.cn



<https://logms2025.shmtu.edu.cn>



+86 15921317091 (Dr. Wang)



Notes

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



Notes

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.